

phase5_rotation_guard.sh — LE Phase-5 renewal §11.4.135 standing guard

Revision: 1 **Last modified:** 2026-07-01T11:40:00Z **Authority:**
Inherits constitution/Constitution.md per §11.4.35. §11.4.18
companion for tests/letsencrypt/phase5_rotation_guard.sh.

Overview

The standing §11.4.135 regression guard for the Let's Encrypt renewal/rotation feature, with a §11.4.115 RED_MODE polarity. It wraps deploy/letsencrypt/phase5_rotation.sh.

- **GREEN (RED_MODE=0, default):** a real renewal rotation (S1→S2, new serial) happens with 0 dropped requests across the swap and the cert-analyzer verifies S2 → **PASS**.
- **RED (RED_MODE=1):** the same run with PHASE5_NO_SURGERY=1 — the deterministic ARI trigger removed — **MUST NOT** renew; phase5_rotation.sh exits non-0 and the guard **PASS**es, proving the ARI-window surgery is what triggers the renewal (not the restart alone). If a renewal happens **WITHOUT** the surgery, the guard **FAIL**s (§11.4.7).

Prerequisites / topology

Boots the hermetic stack (conductor-owned, §11.4.119). 3-way exit — 0=PASS, 2=honest topology SKIP (built image / podman-compose absent, §11.4.3 — never a fake pass), else FAIL. Registered in tests/run-tests.sh test_regression_guards(). Expensive (~1-2 min; boots containers) — set SKIP_LE_ISSUANCE_GUARD=1 to skip the LE guards.

Usage

```
sh tests/letsencrypt/phase5_rotation_guard.sh # GREEN
RED_MODE=1 sh tests/letsencrypt/phase5_rotation_guard.sh # RED
(reproduce: no surgery => no renewal)
```

Runs under GOMAXPROCS=2 nice -n 19 ionice -c 3.

Last verified

2026-07-01 — GREEN PASS (real S1→S2 rotation, 0 dropped, analyzer verifies S2) and RED PASS (with PHASE5_NO_SURGERY=1 no renewal fires → guard catches it). Evidence under qa-results/regression/ + qa-results/letsencrypt/phase5_rotation/.

Related

- [deploy/letsencrypt/phase5_rotation.sh](#) — the proof this guards.
- [tests/letsencrypt/phase3_issuance_guard.sh](#) — the sibling issuance guard.